IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

MODELLING AND CONTROL OF MULTI-BODY MECHANICAL SYSTEMS

Friday, 1 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions

All questions carry equal marks.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S.A. Evangelou
Second Marker(s) : A. Astolfi

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MODELLING AND CONTROL OF MULTIBODY MECHANICAL SYSTEMS

1. A compound pendulum of mass m consists of a symmetric rigid body (shown in Figure 1.1) and a horizontal massless plate (not shown in Figure 1.1). The plate is attached to point O in Figure 1.1 and is free to rotate by an angle ψ about the vertical Earth-fixed axis through O. The rigid body is fixed at one of its end points along its axis of symmetry (point O in Figure 1.1) to the plate and is free to rotate by an angle θ about a horizontal plate-fixed axis through O, under the influence of gravity. The centre of mass of the rigid body is on the axis of symmetry at distance f from point O.

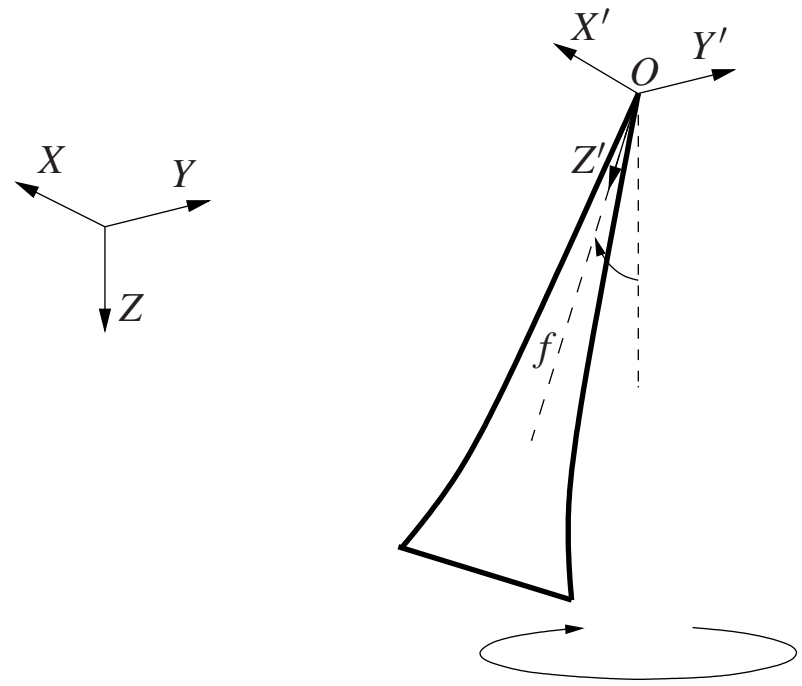


Figure 1.1 Compound 2-degree-of-freedom pendulum.

A body-fixed Cartesian coordinate system has its origin at O, and its z-axis (Z') along the axis of symmetry of the body and its y-axis (Y') along the horizontal plate-fixed axis about which the rigid body rotates by the angle θ. The body-fixed x-, y-, and z-axes ($X'Y'Z'$) correspond to principal axes with principal moments of inertia I_{xx}, I_{yy} $(= I_{xx})$, and I_{zz} respectively. In the nominal configuration the body-fixed x-, y-, and z-axes are respectively aligned with the x-, y-, and z-axes (XYZ) of an Earth-fixed Cartesian coordinate system, in which the x-axis is horizontal and the z-axis points downwards. The angular configuration of the body is therefore given by the Euler angle convention, first rotation from its nominal configuration by angle ψ followed by angle θ rotation.

a) Calculate the body angular velocity vector in the body-fixed coordinate system in terms of the generalised coordinates. [5]

b) Compute the kinetic energy of the compound pendulum. [5]

c) Calculate the potential energy of the compound pendulum, and hence determine the Lagrangian function. [4]

d) Use the Lagrangian approach to derive the equations of motion of the compound pendulum in terms of the generalised coordinates. [6]

2. It is desired to achieve lateral balancing of a humanoid robot by antisymmetric rotation of the robot arms in a vertical plane. As shown in Figure 2.1, the lateral dynamic behaviour of the robot is therefore approximated by a rigid rod OD that is fixed on the ground at point O and is free to rotate about a horizontal longitudinal axis through O by an angle θ, and a uniform rigid rod AB that is attached to the first rod at point C_2 and is free to rotate with respect to the first rod by an angle ϕ about an axis through C_2 that is parallel to the axis of rotation of the first rod, under the influence of gravity. The centre of mass of rod OD is at point C_1 at distance h_1 from O and the centre of mass of rod AB is at point C_2 at distance h_2 from O. The mass and spin inertia (about its centre of mass) of rod OD are m_1 and I_1 respectively, and the corresponding quantities for rod AB are m_2 and I_2 respectively. A control torque T_c acting between the two rods at point C_2 provides the control action for balancing the robot.

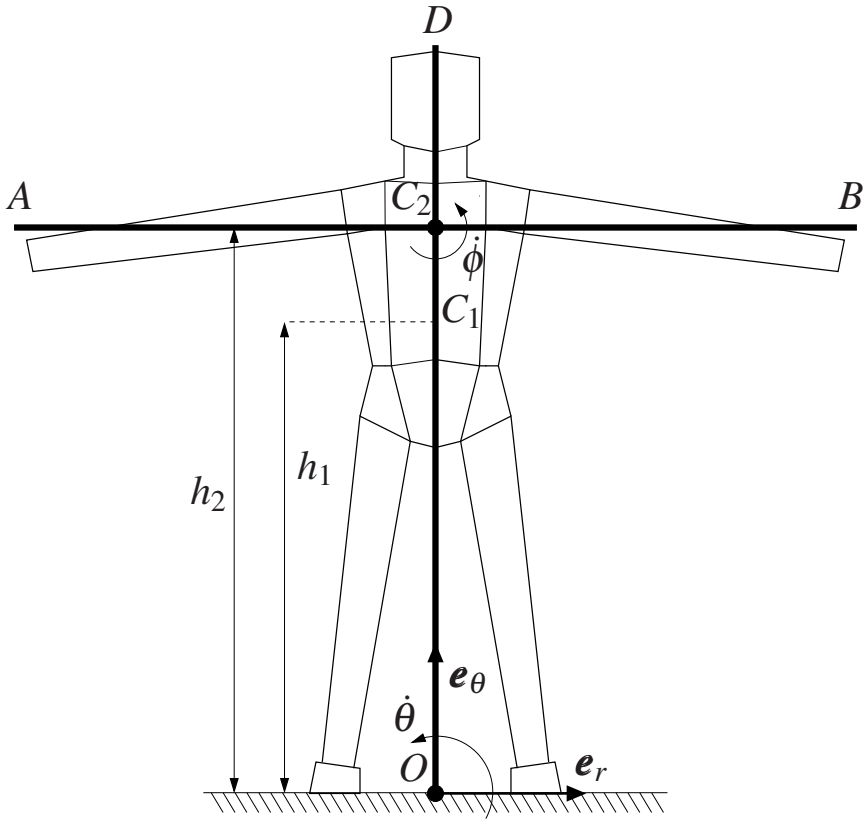


Figure 2.1 Humanoid robot abstraction by two rigid rods.

A moving Cartesian coordinate system with unit vectors $\boldsymbol{e}_r$ and $\boldsymbol{e}_{\boldsymbol{\theta}}$ (see Figure 2.1) is used to analyse the motion of the two rods. The moving coordinate system is attached to the rod OD, it has its origin at O and it rotates by an angle θ. In the configuration shown in Figure 2.1, in which OD is vertical and AB is horizontal, $\theta = 0$ and $\phi = 0$.

a) Write the position and velocity vectors of the centre of mass of each of the two rods. [4]

b) Compute the total kinetic energy of the system. [5]

c) Calculate the total potential energy of the system, and hence determine the Lagrangian function. [4]

d) Use the Lagrangian approach to derive the equations of motion of the system in terms of the generalised coordinates. [7]

3. As in Q1, a compound pendulum of mass m consists of a symmetric rigid body (shown in Figure 3.1) and a horizontal massless plate (not shown in Figure 3.1). The plate is attached to point O in Figure 3.1 and is free to rotate by an angle ψ about the vertical Earth-fixed axis through O. The rigid body is fixed at one of its end points along its axis of symmetry (point O in Figure 3.1) to the plate and is free to rotate by an angle θ about a horizontal plate-fixed axis through O, under the influence of gravity. The centre of mass of the rigid body is on the axis of symmetry at distance f from point O.

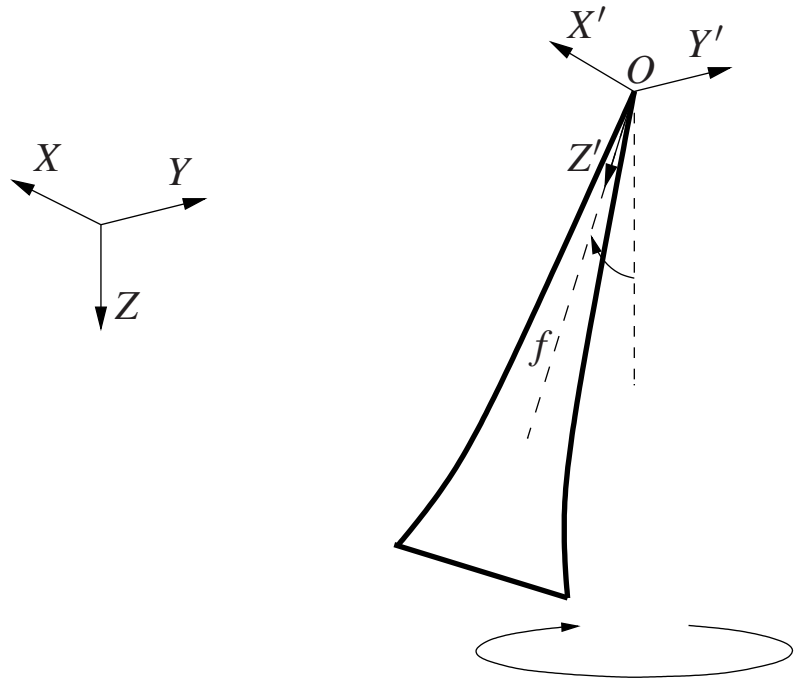


Figure 3.1 Compound 2-degree-of-freedom pendulum.

A body-fixed Cartesian coordinate system has its origin at O, and its z-axis (Z') along the axis of symmetry of the body and its y-axis (Y') along the horizontal plate-fixed axis about which the rigid body rotates by the angle θ. The body-fixed x-, y-, and z-axes ($X'Y'Z'$) correspond to principal axes with principal moments of inertia I_{xx}, I_{yy} $(= I_{xx})$, and I_{zz} respectively. In the nominal configuration the body-fixed x-, y-, and z-axes are respectively aligned with the x-, y-, and z-axes (XYZ) of an Earth-fixed Cartesian coordinate system, in which the x-axis is horizontal and the z-axis points downwards. The angular configuration of the body is therefore given by the Euler angle convention, first rotation from its nominal configuration by angle ψ followed by angle θ rotation.

a) By using the standard single-axis-rotation transformation matrices calculate

   i) the transformation matrix $\boldsymbol{A_1}$ from Earth-fixed to plate-fixed coordinates and the transformation matrix $\boldsymbol{A_2}$ from body-fixed to plate-fixed coordinates, and [3]

   ii) the body angular momentum vector, $\boldsymbol{H}$, in the plate-fixed coordinate system. [5]

b) Compute in the plate-fixed coordinate system the total external moment $\boldsymbol{N}$ acting on the rigid body due to gravity and also in terms of a moment of magnitude M that maintains the rotational constraint of the compound pendulum. (Hint: the compound pendulum has 2 and not 3 rotational degrees of freedom.) [4]

c) By considering the rotational motion about the fixed point O, use the vectorial approach in plate-fixed coordinates to derive

   i) the equations of motion of the compound pendulum, and [4]

   ii) the moment M, in terms of the generalised coordinates. [4]

4. As in Q2, it is desired to achieve lateral balancing of a humanoid robot by antisymmetric rotation of the robot arms in a vertical plane. As shown in Figure 4.1, the lateral dynamic behaviour of the robot is therefore approximated by a rigid rod OD that is fixed on the ground at point O and is free to rotate about a horizontal longitudinal axis through O by an angle θ, and a uniform rigid rod AB that is attached to the first rod at point C_2 and is free to rotate with respect to the first rod by an angle ϕ about an axis through C_2 that is parallel to the axis of rotation of the first rod, under the influence of gravity. The centre of mass of rod OD is at point C_1 at distance h_1 from O and the centre of mass of rod AB is at point C_2 at distance h_2 from O. The mass and spin inertia (about its centre of mass) of rod OD are m_1 and I_1 respectively, and the corresponding quantities for rod AB are m_2 and I_2 respectively. A control torque T_c acting between the two rods at point C_2 provides the control action for balancing the robot.

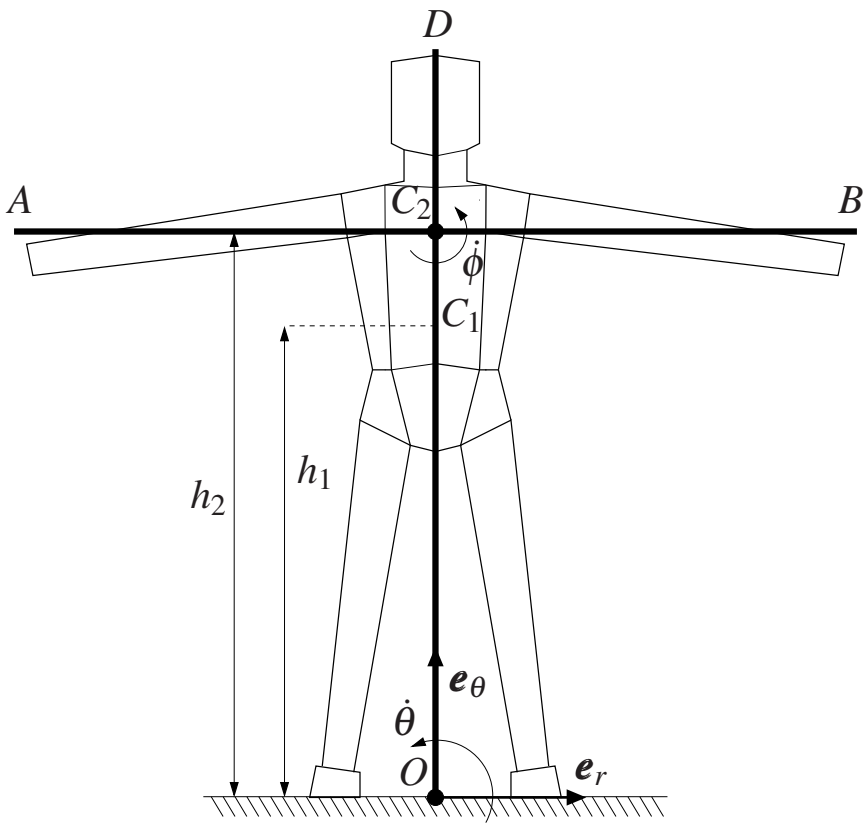


Figure 4.1 Humanoid robot abstraction by two rigid rods.

A moving Cartesian coordinate system with unit vectors $\boldsymbol{e}_r$ and $\boldsymbol{e}_{\boldsymbol{\theta}}$ (see Figure 4.1) is used to analyse the motion of the two rods. The moving coordinate system is attached to the rod OD, it has its origin at O and it rotates by an angle θ. In the configuration shown in Figure 4.1, in which OD is vertical and AB is horizontal, $\theta = 0$ and $\phi = 0$.

a) Determine the generalised coordinates of the system. [2]

b) Write the acceleration vectors of the centre of mass of each of the rods. [3]

c) F_{r1} and $F_{\theta 1}$ are the unknown constraint forces on rod OD from Earth in the $\boldsymbol{e}_r$ and $\boldsymbol{e}_{\boldsymbol{\theta}}$ directions respectively. F_{r2} and $F_{\theta 2}$ are the unknown constraint forces on rod AB from rod OD in the $\boldsymbol{e}_r$ and $\boldsymbol{e}_{\boldsymbol{\theta}}$ directions respectively. Write the total force that acts on each of the rods in vector form. [3]

d) Use the vectorial approach to derive F_{r1}, $F_{\theta 1}$, F_{r2}, and $F_{\theta 2}$ in terms of the generalised coordinates. [6]

e) Use the vectorial approach to further derive the equations of motion of the system with respect to the generalised coordinates. [6]